

VIKAS CHIRRA

Data Analyst

Email: vikasgoud1469@gmail.com | Phone: 9405956537 | Location: United States

PROFESSIONAL SUMMARY

- Detail-oriented **Data Analyst** with over **5 years of professional experience** in analyzing, visualizing, and interpreting large datasets using **SQL, Python, Power BI, and Tableau** to deliver actionable business insights and data-driven recommendations.
- Expertise in **data extraction, transformation, and loading (ETL)** using tools such as **SSIS, Azure Data Factory, and Informatica**, ensuring seamless integration between diverse data sources for accurate analytics.
- Proficient in **SQL query optimization**, joins, stored procedures, and views creation to improve performance and deliver reliable datasets for **enterprise reporting and business intelligence solutions**.
- Skilled in building **interactive dashboards** using **Power BI and Tableau**, translating complex KPIs into visually appealing visualizations that drive executive-level decisions across finance, marketing, and operations.
- Experienced in **data preprocessing, cleaning, and feature engineering** using **Python libraries** such as **Pandas, NumPy, and Scikit-learn**, ensuring data consistency and readiness for advanced analytics.
- Strong understanding of **data warehousing, dimensional modeling (star and snowflake schema), and fact-dimension relationships** to support analytical queries and reporting layers.
- Hands-on experience in **automating repetitive reporting tasks** using **Python scripting, Excel VBA, and Power Automate**, reducing manual efforts and improving operational efficiency.
- Expertise in **data validation and quality checks** to ensure accuracy, consistency, and completeness of data before analytics and visualization stages.
- Proficient in leveraging **Excel advanced functions**, including **Pivot Tables, Power Query, Index-Match, and VBA Macros**, for trend analysis and business performance monitoring.
- Collaborated with **cross-functional teams** to gather business requirements, design analytical frameworks, and deliver scalable insights supporting both operational and strategic decisions.
- Designed and deployed **predictive analytics models** to identify trends, detect anomalies, and forecast sales or customer behavior using **regression, classification, and clustering techniques**.
- Adept at **data storytelling**, creating impactful narratives with visual data that clearly communicates trends, risks, and opportunities to non-technical stakeholders.
- Familiar with **cloud-based analytics** using **Azure Synapse, Snowflake, and AWS Redshift**, managing datasets stored in cloud storage environments with high security and accessibility.
- Ensured **data governance, compliance, and security** by adhering to enterprise standards, access control policies, and privacy guidelines like **GDPR and HIPAA**.
- Strong foundation in **statistical analysis** using descriptive and inferential techniques to support hypothesis testing, customer segmentation, and marketing campaign performance measurement.
- Experienced in **Agile project delivery**, actively participating in sprint planning, retrospectives, and backlog prioritization to align analytics goals with business timelines.
- Worked closely with business teams to **translate vague requirements into measurable KPIs**, dashboards, and reports, ensuring end-user satisfaction and operational excellence.
- Continuously focused on **improving process efficiency**, adopting new technologies, and upskilling in **data visualization, automation, and AI-driven analytics tools** to stay ahead of industry trends.

TECHNICAL SKILLS

- **Data Visualization:** Power BI, Tableau, Excel Power Pivot
- **Programming Languages:** Python, SQL, T-SQL, DAX, R
- **ETL & Data Integration:** SSIS, Azure Data Factory, Alteryx
- **MDM Tools:** Master Data Management
- **Databases:** SQL Server, Azure SQL, Snowflake, MySQL

- **Data Analysis Tools:** Pandas, NumPy, Matplotlib, Seaborn
- **Cloud Platforms:** Azure Synapse, Databricks, Microsoft Dynamic 365, AWS Redshift, Google BigQuery, GCP
- **Excel Skills:** Power Query, Macros, Pivot Tables, VLOOKUP
- **Methodologies:** Agile, Scrum, Data Warehousing, Data Modeling, Jira

PROFESSIONAL EXPERIENCE

Client: Providence Health & Services | Role: Data Analyst | Duration: Aug 2024 - Present

- Designed and automated **ETL pipelines** using **Azure Data Factory** to extract and process healthcare data from multiple EMR and billing systems, ensuring consistent data flow into **Azure Data Lake Storage**.
- Reviewed healthcare vendor agreements, data sharing contracts, and service level agreements (**SLAs**) to ensure compliance with **HIPAA**, data governance standards, and operational requirements.
- Analyzed payer and provider contract terms to validate **reimbursement logic, performance metrics, and reporting** requirements using claims and revenue cycle data.
- Designed and implemented data visualizations for key business metrics, enabling management to monitor performance and make data-driven decisions.
- Built complex reports and **KPIs** using drag-and-drop visualization platforms and **SQL** based data sources, comparable to **QlikView/Qlik Sense** functionality.
- Created **Power BI dashboards** integrating patient, clinical, and operational data to help leadership identify resource gaps, improve patient flow, and optimize hospital occupancy rates.
- Developed and maintained **KPI dashboards and performance reports** using Power BI and Tableau, delivering actionable insights to stakeholders, experience is easily transferable to tools like **Sigma or Mode**.
- Analyzed **sales performance, order fulfillment, and product profitability**, providing insights to optimize operations and support supply chain decision-making; familiar with logistics workflows and inventory management processes.
- Gained hands-on experience with **Databricks** on Azure, developing and executing **PySpark** and **Spark SQL** notebooks to process large-scale datasets, perform transformations, and optimize distributed data workflows for analytics use cases.
- Built and tested end-to-end distributed data pipelines using PySpark, leveraging Spark SQL for aggregations and performance tuning, while integrating cloud storage **Azure Data Lake / AWS S3** to support scalable, cloud-native data processing.
- Utilized **Python (Pandas, NumPy)** for advanced data wrangling, cleansing, and transformation to prepare structured datasets suitable for predictive analytics and performance monitoring.
- Implemented **data quality validation rules** using **SQL and ADF data flows**, ensuring accurate and reliable reporting data across multiple business domains including finance and HR.
- Partnered with medical and financial teams to define **KPI metrics** and automated performance reports, reducing manual reporting efforts by 60% across departments.
- Performed **root-cause analysis** on data discrepancies between clinical and finance systems, collaborating with DBAs and data engineers to optimize query logic and improve data consistency.
- Built **interactive dashboards** using **Power BI DAX measures** and complex data modeling techniques, allowing leadership to visualize patient outcomes, cost variances, and operational trends in real time.
- Developed **automated Python scripts** for daily data quality validation, exception handling, and alerting processes, minimizing manual intervention and ensuring timely data delivery.
- Conducted **statistical analysis and forecasting** using regression and time-series methods in Python to predict patient admissions and seasonal utilization trends.
- Integrated multiple data sources including **EPIC, Oracle EBS, and SQL Server**, building a unified analytical environment for organization-wide insights.
- Optimized **SQL queries and stored procedures** for faster data retrieval, ensuring efficient Power BI refresh rates and improved dashboard responsiveness.
- Documented **data lineage, transformation rules, and business logic** for audit and compliance purposes, maintaining transparency in healthcare analytics processes.
- Collaborated with **data governance teams** to implement access control, role-based permissions, and compliance with HIPAA data privacy regulations.

- Conducted regular **data validation checks and reconciliation** between ADF pipelines and Power BI datasets to maintain integrity in executive reporting.
- Coordinated with cross-functional stakeholders including **doctors, analysts, and IT managers** to design analytics that improved patient care metrics and operational planning.
- Designed **performance optimization strategies** for ETL jobs and Power BI data models, improving pipeline efficiency by 30% and ensuring uninterrupted dashboard availability.

Client: Key Bank | Role: Data Analyst | Duration: May 2021 – Jul 2023

- Designed **SQL-based data models** and optimized queries to analyze financial transactions, loan performance, and customer risk profiles, supporting regulatory and compliance reporting initiatives.
- Built **Power BI dashboards** visualizing key banking metrics such as loan default trends, credit exposure, and profitability across different portfolios.
- Collaborated with data engineering teams to **develop ETL pipelines in Azure Synapse and Data Factory**, consolidating structured and unstructured data from multiple banking applications.
- Developed **Python scripts for data validation, cleansing, and enrichment**, automating routine workflows and minimizing human errors in analytics processes.
- Applied **statistical and predictive modeling techniques** to forecast credit risk and customer retention probabilities, aiding risk management and marketing strategy decisions.
- Implemented **data quality monitoring dashboards** that automatically flagged inconsistencies in transaction and account-level data for proactive correction.
- Integrated external market data APIs with internal banking systems to enable real-time analytics on customer demographics and lending risk factors.
- Built **advanced DAX calculations** in Power BI for financial ratios, risk-weighted assets, and net interest margin analysis used by senior executives.
- Worked closely with auditors to ensure **SOX and Basel III compliance** by validating accuracy in all regulatory reporting datasets.
- Managed **ad-hoc data requests** from compliance, treasury, and lending teams by creating SQL-based extracts and Power BI visuals for trend analysis.
- Designed **data lineage documentation** to trace the origin and transformation of critical datasets across the data warehouse.
- Developed **interactive Tableau dashboards** for visual comparison of loan approval trends and repayment patterns across customer segments.
- Automated **monthly and quarterly business review reports** using Excel VBA and Power BI scheduling, reducing reporting turnaround time by 40%.
- Partnered with IT and business users to define **metadata standards, naming conventions, and business glossary** for consistent analytics.
- Conducted **performance tuning** on SQL queries and cube models, improving data retrieval speed and overall Power BI responsiveness.
- Supported **data migration activities** during banking application modernization, ensuring legacy datasets were mapped correctly to new systems with zero data loss.

Client: Newegg | Role: Data Analyst | Duration: Jan 2019 – May 2021

- Designed and maintained **sales performance dashboards** using Power BI and Excel to track product profitability, order fulfillment, and regional sales trends across e-commerce operations.
- Utilized **SQL and Python scripts** to analyze customer purchase behavior, identify seasonal demand spikes, and recommend product pricing adjustments to improve revenue.
- Developed **data extraction workflows** integrating data from ERP, CRM, and Google Analytics into centralized SQL databases for unified reporting.
- Built **KPI-based scorecards** for marketing campaigns using Power BI DAX calculations and visual storytelling techniques for executive-level presentations.
- Implemented **data validation processes** using SQL triggers and stored procedures to ensure accuracy in daily sales and inventory reports.

- Applied **predictive modeling in Python** to forecast product demand, optimize inventory levels, and minimize stockouts or overstocking situations.
- Created **automated ETL pipelines** to ingest product, customer, and transaction data into Azure SQL databases, enabling near real-time analytics.
- Conducted **A/B testing and performance analysis** for promotional campaigns using Python statistical libraries, quantifying conversion rate improvements and ROI.
- Collaborated with product, supply chain, and marketing teams to design actionable insights that improved online customer experience and marketing efficiency.
- Built **time-series forecasting models** to predict daily sales, using ARIMA and exponential smoothing techniques for better inventory management.
- Designed **data-driven marketing dashboards** integrating Google Analytics, AdWords, and Salesforce data for campaign optimization.
- Ensured **data integrity and reconciliation** between transaction databases, warehouse systems, and online storefronts through SQL-based validation scripts.
- Automated **data refreshes and report delivery** using Power BI Service and Excel macros, ensuring decision-makers received timely, updated analytics.
- Partnered with engineering teams to improve **data warehouse schema design**, enhancing query performance and analytical flexibility.
- Conducted detailed **profit margin and cost analysis** using Excel Power Query and DAX measures to identify underperforming product categories.
- Contributed to **data governance and documentation efforts**, standardizing data definitions and KPIs for consistency across analytics and reporting teams.

EDUCATION

Masters in Data Science

University of North Texas – Denton, Texas

CERTIFICATION

- Microsoft Certified: Data Analyst Associate (Power BI)
- Microsoft Azure Fundamentals (AZ-900)
- Google Data Analytics Professional Certificate
- Azure Data Fundamentals (DP-900)